UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2011 question paper for the guidance of teachers

5014 ENVIRONMENTAL MANAGEMENT

5014/21 Paper 2, maximum raw mark 60

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1 (a) (i) *fish*: protein/oils/omega 3 and or 6/vitamin A/vitamin D/calcium;
rice: carbohydrate/energy source/vitamin B/fibre;
(do not allow vitamins/minerals/nutrients unqualified) [2]

(ii) so can still produce food/rice during flooding;
grows as fast as water rises/eq;
no need to try and control water level/eq;
harvest earlier;
(any two) [2]

(iii) table that can record the data;
suitable headings;;
(must be for 7 days and a minimum of one of farmer/catch/number/type/weight) [3]

(iv) reliability/eq;
to calculate average/mean;
so catches can be compared;/to give a more complete picture;
(any one)
(do not allow precise/accurate) [1]

(b) (i) weight;
size/length;
sex;
age;
(do not allow number/type)
abiotic factor e.g. temp/pH/specific pollutant/time spent fishing;
(any two) [2]

(ii) keep taking samples for more than one year/eq;
in exactly the same way/eq;
compare the catches/eq;
ref to LEDC's/eq;
(any two) [2]

(c) (i) A; [1]

(ii) C; [1]

(iii) overfishing elsewhere/more people fishing;
from same village; pollution;
valid example of pollution;
fish disease;
(any two) [2]

(d) kills all fish;
kills young fish;
food chains altered;
further appropriate explanation;
not breeding fast enough;
no breeding stock left;
many fish not used for food/eq e.g. left in water;
ref to eutrophication;
(any three) [3]

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(e) (i) licences;
fishing patrols;
fines/eq;
quotas;
set mesh size;
close season;
no catch areas/eq;
AVP;
(any two) [2]

(ii) EITHER
yes: makes people aware; may help to save habitat; saved by captive breeding; more funding; more scientific interest; ref to law; (any two)

OR
no: people will keep capturing it; habitat cannot be saved/eq; so few left its likely to become extinct/eq; more desirable; more poaching; increased price of fish; (any two) [2]

(f) (i) still able to fish;
some power;
still some silt;
less annual flooding;
still able to grow rice;
ref to generating local jobs;
(any three) [3]

(ii) more electricity/energy (than B);
for expanding population;
more flood control;
more aid to build dam/eq;
more jobs (than B);
more money made by government (more quickly);
economy of scale;
ref to developing more industry;
(any two) [2]

2 (a) (i) some correct working; most C; least B; [3]

(ii) orientation;
axes labelled (weight/kg and year/time or all years labelled);;
plots (no errors);
(allow line or bar graphs) [4]

(iii) 2005/2008; (allow ref to third or sixth year) [1]

(iv) 26.25; × 10 = 262.5 (262/263);
OR 262.5 (262/263);; [2]

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(b) (i) hot/warm/tropical;
high rainfall/wet/monsoon;
free standing water;
(allow savanna)
(any two) [2]

(ii) rapid breeding of mosquito/eq;
mosquito's breed in water;
too much water;
cannot all be drained;
as needed for rice;
cannot spray all the water;
so not all mosquitoes can be killed;
no vaccine yet;
ref to cost;
cannot stop mosquitoes biting humans;
(any three) [3]

(iii) make a repellent from cashew shell/gum;
use of a repellent;
apply at night;
use of nets;
keep covered (at night);
ref to reducing breeding sites of mosquitoes;
anti malarial drugs;
(any two) [2]

3 (a) (i) October/November to March; [1]

(ii) any three valid jobs;;
(3:2 2:1) [2]

(b) (i) any three valid questions such as income, employment, number of dependents, relevant work skills ;;;
layout; [4]

(ii) allowed developments: such as sewage works; small shops/hotels/lodges/camps; small roads; paths for hiking; more jobs; other valid ideas; (max 4)

Restrictions: such as large hotels; restaurants; golf course; large roads; dumping wastes; other valid ideas; (max 4) [8]
(Look for suitable qualifications in both parts/')

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